Assignment 01: TSID for bipedal applications

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1 Weights and Gains in TSID

CoM position shows minor deviations: x within 1 mm, y tracks the reference but undershoots peaks, z oscillates $\sim 10^{-4}$. Velocity and acceleration follow similar trends: y tracks but misses peaks, x and z oscillate $\sim 10^{-4}$–10^{-3}, except z acceleration reaching $\sim 10^{-2}$. From the acceleration plots, two issues are evident:

- **Reference trajectory compromise**: The actual trajectory (blue) deviates from the desired one (green), indicating that the CoM acceleration $\ddot{q}_{\text{CoM}}$ has limited influence on the control decision. This behavior is primarily attributed to the low task weight w_{CoM} assigned to the center-of-mass objective.
- **Under-performance**: Peak tracking errors between the actual and reference signals, indicate an insufficient proportional gain k_p.

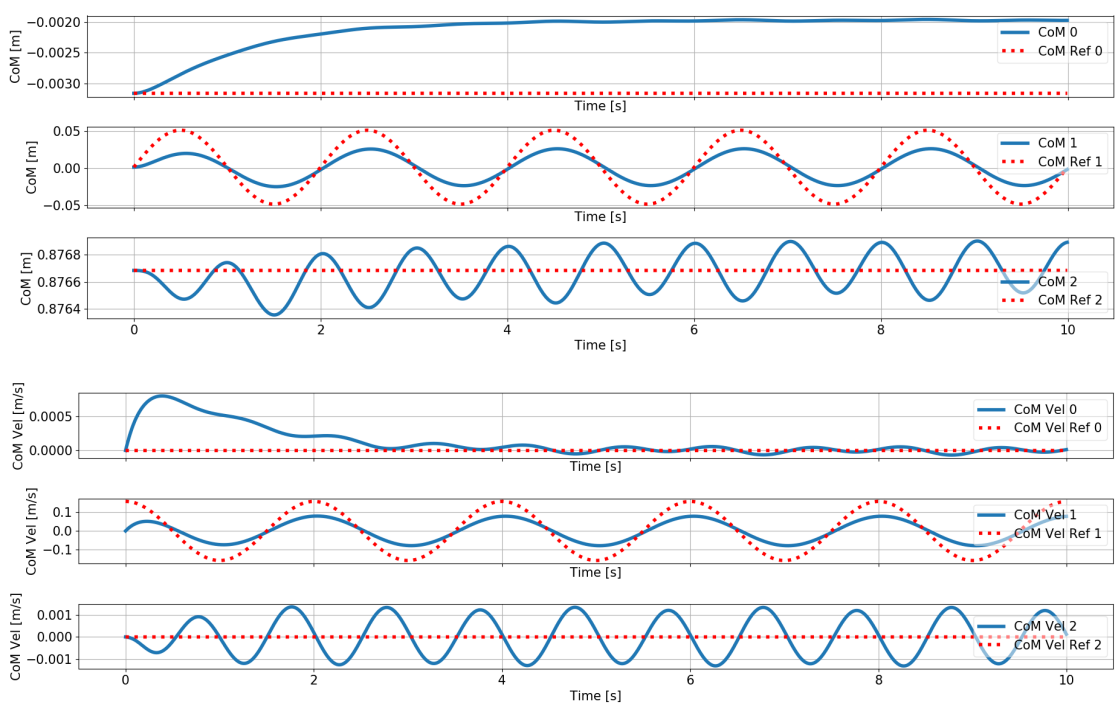

Figure 1: $w_{\text{CoM}} = 1$, $k_p = 10$ Position and Velocity Tracking

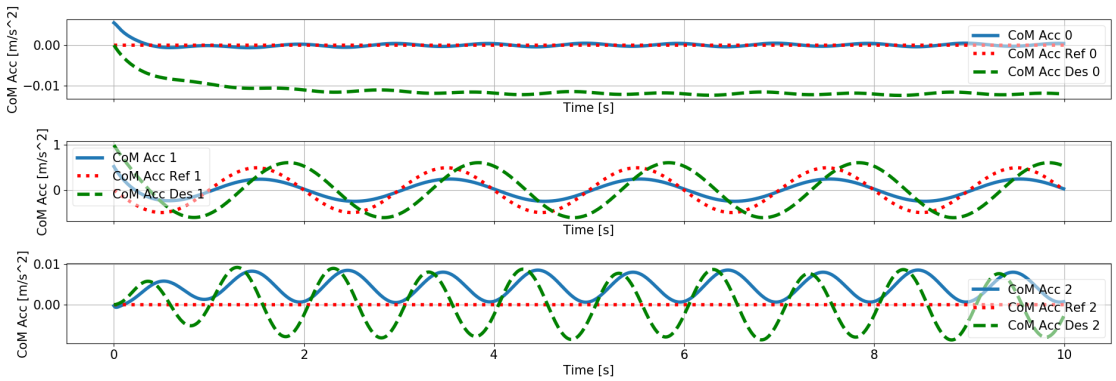

Figure 2: $w_{\text{CoM}} = 1$, $k_p = 10$ Acceleration Tracking

1.1 Effect of Modifying Weights and Gains

Two strategies were explored to enhance CoM tracking: increasing the task weight w_{CoM} to raise its priority, or increasing the proportional gain k_p to strengthen tracking effort.

Config 1, increase the task relevance: $w_{\text{CoM}} = 10$, $k_p = 10$.

Config 2, increase the task magnitude: $w_{\text{CoM}} = 1$, $k_p = 100$.

Config	w_{CoM}	k_p	$\text{RMSE}_{\text{overall}}$	Std_{x}	Std_{y}	Std_{z}
0	1	10	1.09×10^{-1}	6.03×10^{-4}	1.89×10^{-1}	2.68×10^{-3}
1	10	10	5.78×10^{-2}	8.72×10^{-4}	9.79×10^{-2}	6.11×10^{-3}
2	1	100	7.89×10^{-2}	1.38×10^{-3}	1.35×10^{-1}	1.16×10^{-2}

Table 1: Performance metrics on acceleration trajectory tracking

Both configurations improve trajectory tracking, especially along the y-axis (main motion), but also introduce stronger oscillations in x and z, reflected by higher standard deviations (Std_{x}, Std_{z}) in Table **??**.

In Config 1, characterized by a higher task weight (w_{CoM}=10) and moderate gain (k_p=10), the system achieves superior tracking accuracy and reduced oscillations in the transverse directions. However, it exhibits a slower response, requiring more time to converge to the reference trajectory, as highlighted in the plots.

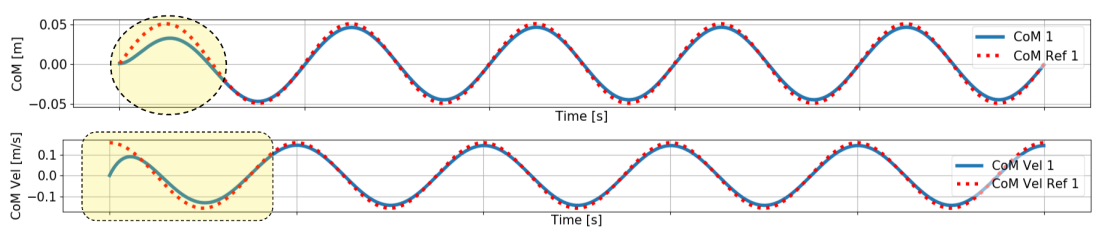

Figure 3: $w_{\text{CoM}} = 10$, $k_p = 10$ Position and Velocity y Axis Tracking

In contrast, Config 2 (w_{CoM}=1, k_p=100) prioritizes reactivity: the higher gain enables a faster response to small errors in position and/or velocity, which brings about a quicker convergence to the desired trajectory. This comes at the cost of slightly degraded tracking performance and increased oscillatory behavior, a consequence of the controller's aggressive action.

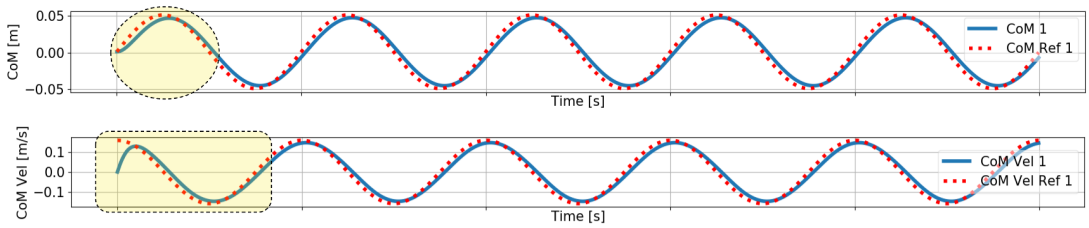

Figure 4: $w_{\text{CoM}} = 1$, $k_p = 100$ Position and Velocity y Axis Tracking

These results suggest that Config 1 is better suited for general, predictable scenarios where accuracy and smoothness are prioritized, whereas Config 2 may be preferable in highly dynamic or unpredictable environments where rapid reaction is critical.

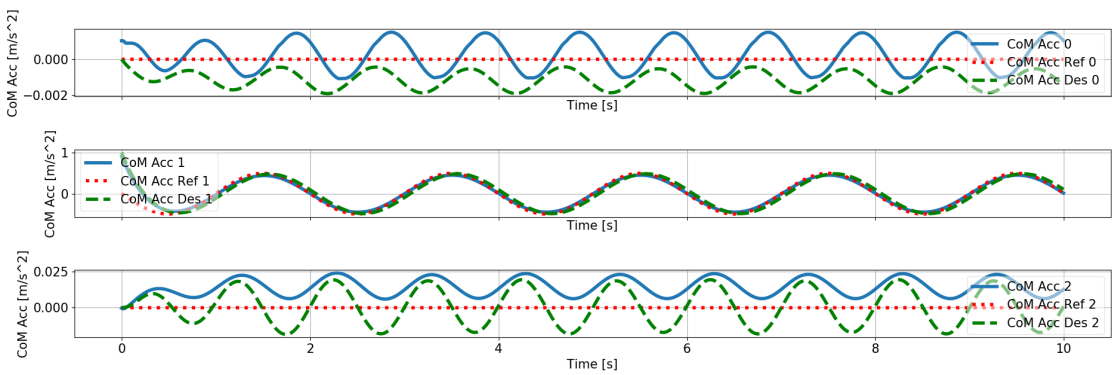

Figure 5: $w_{\text{CoM}} = 10$, $k_p = 10$ Acceleration Tracking

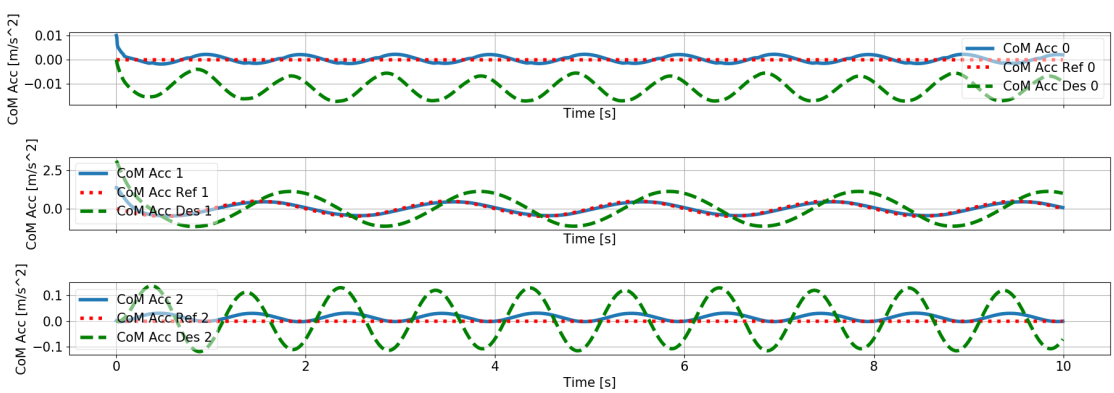

Figure 6: $w_{\text{CoM}} = 1$, $k_p = 100$ Acceleration Tracking

1.2 Higher Frequency Trajectory

In both scenarios, the robot struggles to track the desired trajectory. Higher reference frequency demands faster responses and greater joint accelerations. In both cases, actuators reach their acceleration limits while attempting to track the faster trajectory, as shown by flat regions in the acceleration plots.

In Config 1, the lower gain delays error compensation, leading to cumulative deviations, amplified CoM errors, and eventually saturation of joint accelerations.

Conversely, Config 2, which employs a higher gain, enables more accurate trajectory tracking. Despite this improvement, an initial delay persists, indicating that the controller operates near its performance limits.

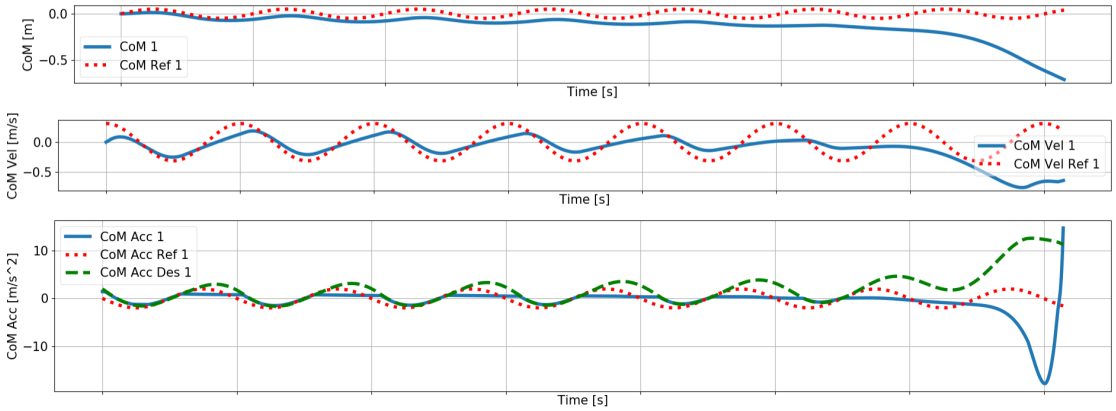

Figure 7: $w_{\text{CoM}} = 10$, $k_p = 10$ Position, Velocity and Acceleration y Axis Tracking

A notable observation is the asymmetric behavior between the left and right sides of the robot. As the robot initially moves to the left, larger errors develop on the opposite side, which the controller attempts to correct by generating higher desired accelerations in the positive direction.

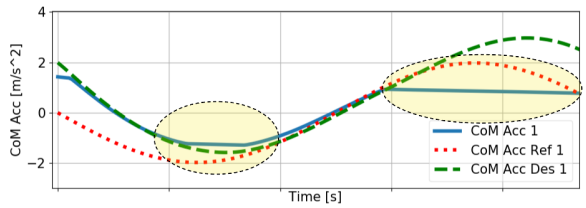

Figure 8: Acceleration y Axis Zoom.

The full amplitude of the reference trajectory is not achieved due to acceleration saturation. Overall, the higher-gain configuration provides better tracking accuracy and improved stability.

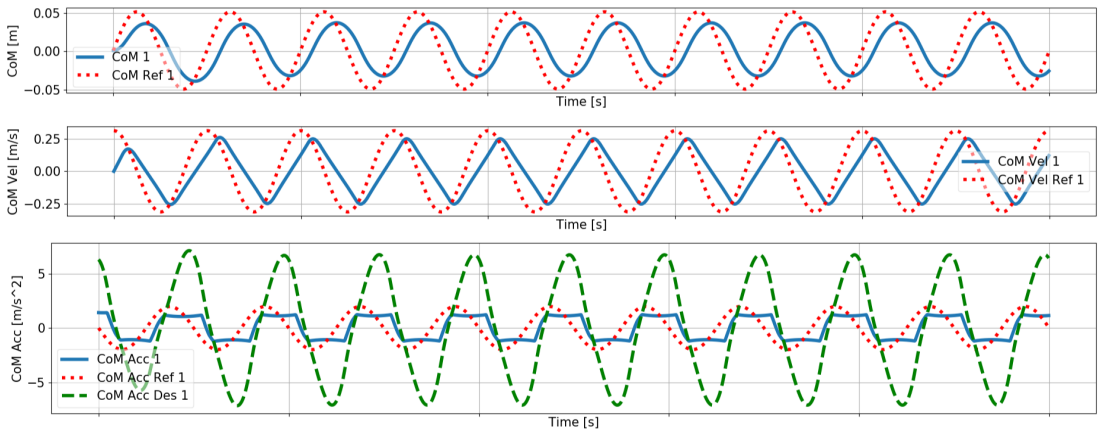

Figure 9: $w_{\text{CoM}} = 1$, $k_p = 100$ Position, Velocity and Acceleration y Axis Tracking

2 Importance of tasks for biped walking

2.1 Step's Length 15

Postural Task Weight (w_{posture}): This parameter allows the robot to maintain an upright posture. When $w_{\text{posture}} = 0$, the robot collapses immediately. For very small values, the robot cannot stay upright, whereas excessively large values deteriorate other task performance. Two configurations were found to be effective:

1. $w_{\text{posture}} = 1 \times 10^{-3}$: excellent CoM tracking, but the two knees joints reach their torque limits.
2. $w_{\text{posture}} = 1 \times 10^{-1}$: slightly worse CoM tracking, but not reached torque limit.

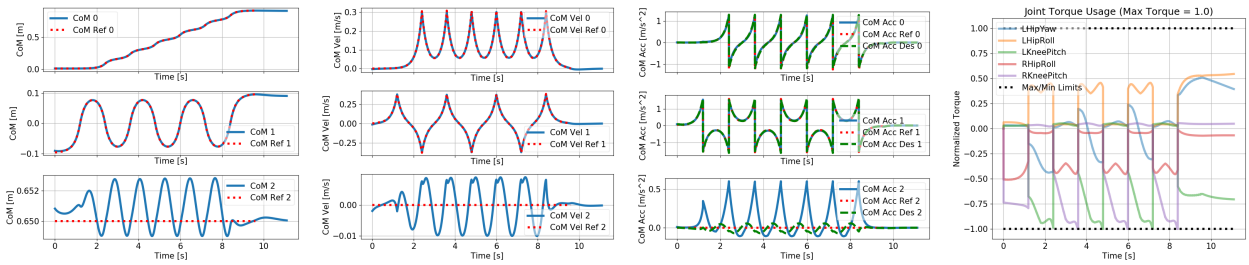

Figure 10: $w_{\text{posture}} = 1 \times 10^{-3}$ Position, Velocity, Acceleration and Joints Torques.

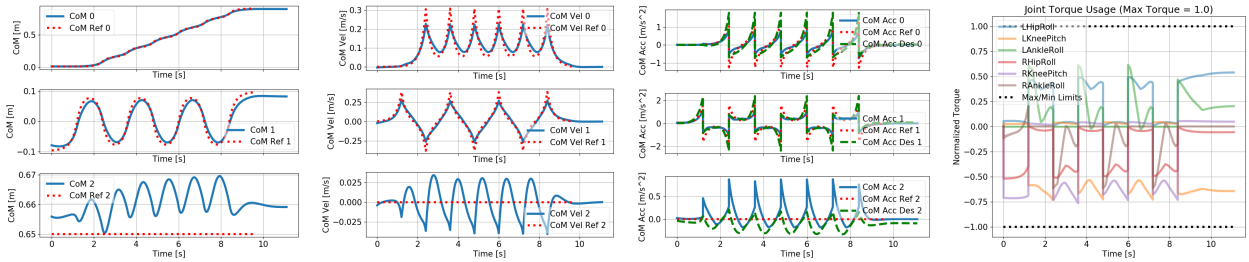

Figure 11: $w_{\text{posture}} = 1 \times 10^{-1}$ Position, Velocity, Acceleration and Joints Torques.

CoM Task Weight (w_{com}): This weight determines the importance of the CoM trajectory tracking.

3. $w_{\text{posture}} = 10^{-3}$, $w_{\text{com}} = 0.1$: lowering w_{com} slightly degraded CoM tracking without reducing joint torque issues, not so effective.
4. $w_{\text{posture}} = 10^{-1}$, $w_{\text{com}} = 10$: increasing w_{com} improved CoM tracking compared to case 2 without torque limit violations. However, this configuration presents two drawbacks: It increases all task weights, which is not recommended in control design. Foot trajectory tracking significantly worsens.

Table 2: Summary of configuration performance metrics

Config	RMSE x pos	RMSE y pos	RMSE x vel	RMSE y vel	RMSE x acc	RMSE y acc	R Knee Normalized Torque	L Knee Normalized Torque
1	5.22e-4	9.57e-4	2.48e-3	3.34e-3	3.88e-2	3.07e-2	-1.00	-1.00
2	6.10e-3	1.30e-2	2.75e-2	3.89e-2	2.19e-1	2.78e-1	-0.73	-0.76
3	3.35e-3	6.57e-3	1.66e-2	2.06e-2	1.45e-1	1.46e-1	-0.99	-1.00
4	8.04e-4	2.17e-3	3.82e-3	6.93e-3	3.71e-2	5.73e-2	-0.87	-0.93

Foot Task Weight ($w_{\mathbf{foot}}$): This weight governs the accuracy of the reference foot trajectory. When w_{foot} is too small, the robot drags its feet instead of lifting them. Increasing w_{foot} improves foot trajectory tracking but degrades the accuracy of the initial foot position, which diverges from the reference.

5. Starting from configuration (4), w_{foot} was increased from 1 to 10 to improve foot tracking. Although accuracy improved—particularly along the z-axis—joint torques again reached their limits.This confirms that excessively increasing all task weights degrades overall performance and stability.

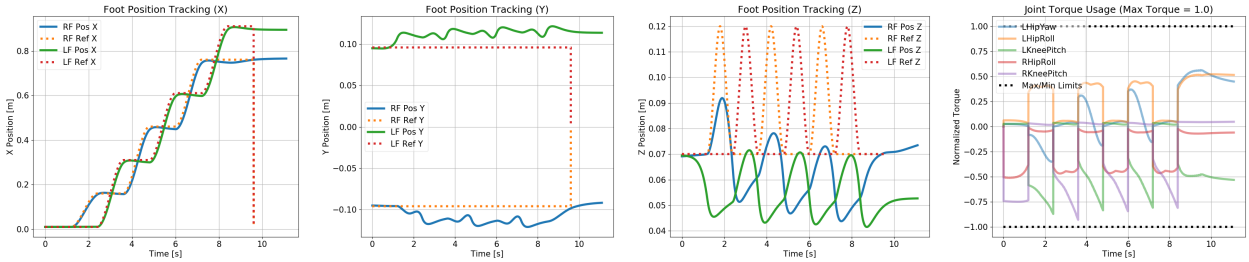

Figure 12: $w_{\text{foot}} = 1$ Foot Trajectory x,y,z and Joints Torques.

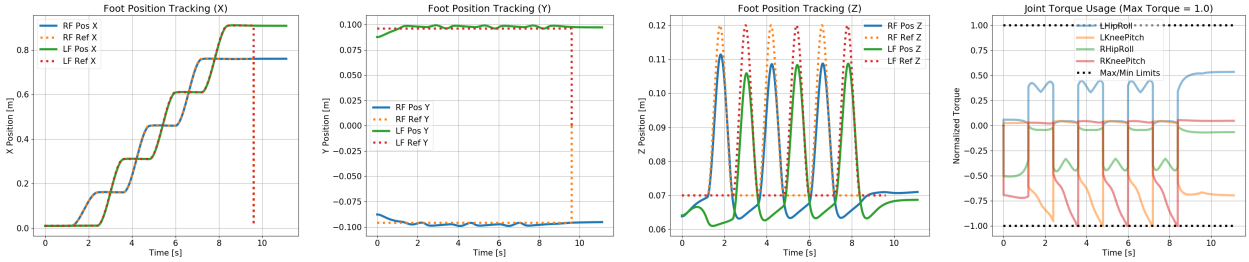

Figure 13: $w_{\text{foot}} = 10$ Foot Trajectory x,y,z and Joints Torques.

Since $w_{\text{posture}} = 10^{-1}$ proved ineffective, we reverted to configuration (1) to address the joint torque limits. Of all weights tested, only the Angular Momentum weight (w_{am}) significantly affected the knees joints torques.

Angular Momentum Weight ($w_{\mathbf{am}}$): This parameter regulates the angular momentum of the CoM, thereby stabilizing the upper body, and maintaining the center of pressure closer to the centers of the feet. If too small, the upper body moves with the lower limbs; if too large, the system becomes unstable.

6. Starting from config (1), varying w_{am} showed that $w_{\text{am}} = 1 \times 10^{-4}$ effectively reduced knee torque saturation (96–97% of the torque limits) while maintaining stability. Increasing it further decreased torque saturation but introduced visible instability, as confirmed by simulation data and visual inspection.

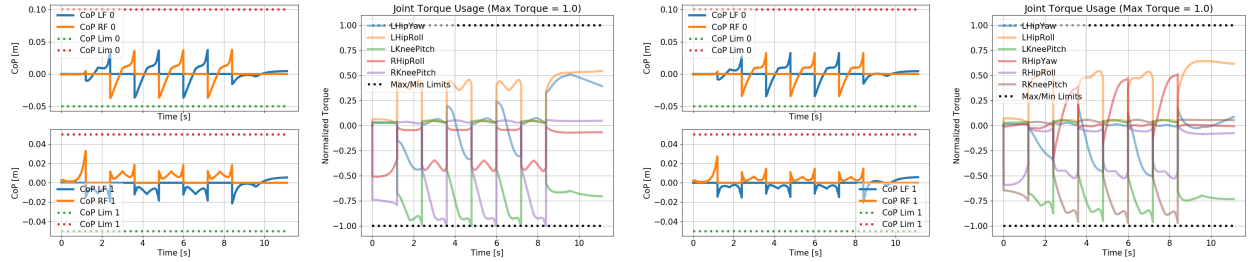

Figure 14: CoP and Joints Torue plots for Config (1) and (6)

To understand this behavior, we compared screenshots of config (1) at the instants when it reach the torque limits with those taken at the same instant for config (6). Visual inspection suggests that the upper body's inertial motion reduces the acceleration required at the knees, mitigating torque peaks.

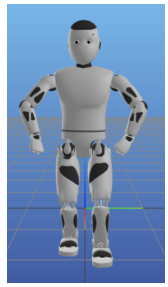

Figure 15: Screenshot Config(1)

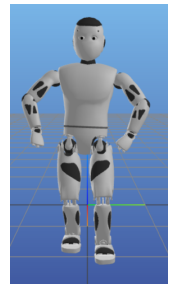

Figure 16: Screenshot Config(6)

Foot Task Weight (w_{foot}): After addressing the torque issue, we come back to tuning w_{foot}

7. Starting from configuration (6), which tracked the feet well but missed z-axis peaks, we increased w_{foot} from 1 to 10. This improved overall tracking but increased the initial position error, causing larger deviation from the reference at the start.

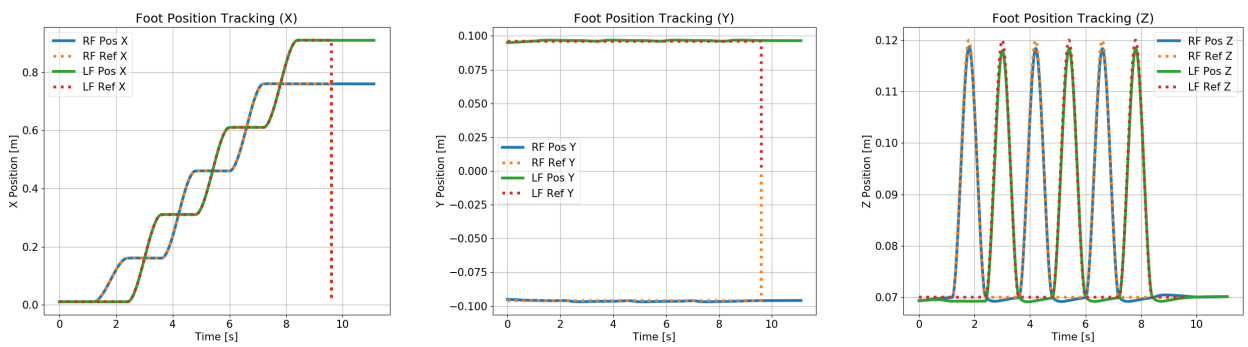

Figure 17: Config (6) $w_{\text{foot}} = 1$ Foot Trajectory x,y,z.

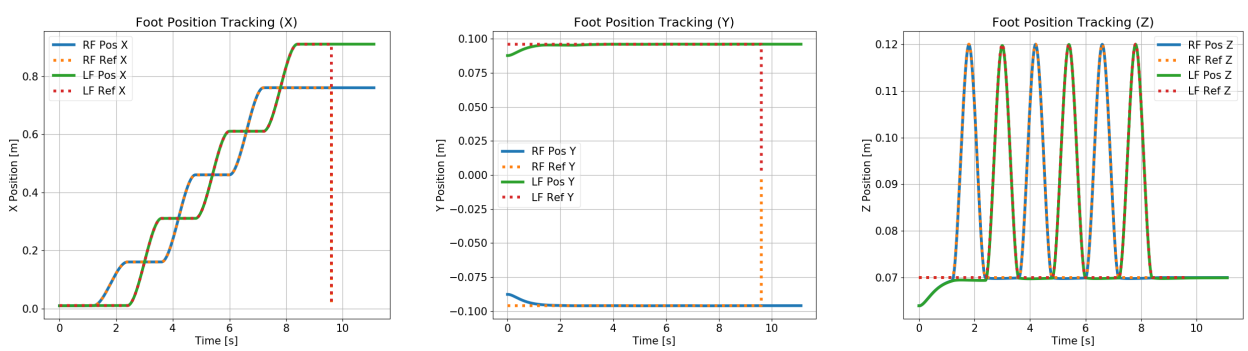

Figure 18: Config (7) $w_{\text{foot}} = 10$ Foot Trajectory x,y,z.

Contact Task Weight (w_{contact}): This weight regulates the importance of maintaining foot–ground contact. Excessively large values divert control effort from other tasks (such as CoM tracking), while too small values allow excessive foot motion, potentially leading to instability or loss of contact.

8. To correct the initial deviation in the foot trajectories, we increased w_{contact} to 10^3, which provided a good balance between foot placement accuracy and torque minimization.

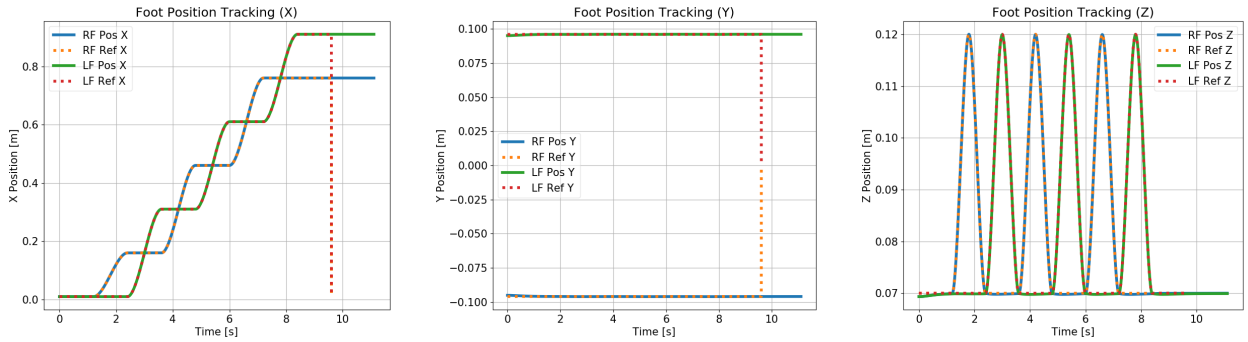

Figure 19: Config (8) $w_{\text{contact}} = 1e3$ Foot Trajectory x,y,z.

2.2 Step's Length 30

In the previous section, we analyzed the impact of each weight on walking with 15cm steps. In the following section, we'll analyze how we had to adjust the weights to allow the robot to walk with 30cm steps.

After some experiments modifying the weights individually, we found that the best configuration to start with was to scale all weights by 10^2 (experiment 1), maintaining the balance found in the previous section but instructing the controller to apply more torque to all joints. We started with this default configuration and repeated the same experiments we had performed for the 15cm steps (scaling everything by 10^2), and the main differences were as follows.

By decreasing **w_posture** (experiment 2), we first achieved better tracking of the COM, without any particular drawbacks. The system was slightly more unstable but did not compromise walking. With 30cm steps, it's true that the COM initially tracks more precisely, but the system instability due to a too-low **w_posture** is much more impactful in this case, causing the robot to fall after a few steps. Even in subsequent experiments, such as Experiment 4, we noted that if **w_posture** is not high enough, the robot cannot remain upright.

Returning to the optimal configuration for **w_posture**, we tried increasing **w_com** (Experiment 3). In this case, we found significantly better tracking than the COM, at the expense of poorer tracking of the foot trajectory and the maximum torques applied for a longer period.

In subsequent experiments, we played with other parameters such as **w_am**, **w_foot**, and **w_contact**, finding that the most important thing for the robot to walk well is that the various weights are balanced. Increasing **w_foot** without correcting **w_am** (experiment 5) makes the robot very unstable, correcting **w_am** and varying **w_foot** (experiments 6 and 7) results in significantly better tracking.